Zhongyi Tong

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EDUCATION

Worcester Polytechnic Institute (WPI) Worcester, MA

Bachelor of Science in Computer Science Aug 2024 - May 2027

GPA: 3.78/4.0

RESEARCH EXPERIENCE

Research Assistant | CVMC Lab, University of Texas at Dallas

Advisor: Prof. Yapeng Tian

- Developed **Harmony**, an agentic audio editing framework integrating LLM planning, audio understanding, and specialized tools for sound addition, removal, replacement, and volume adjustment.
- Designed evidence-guided planning and execution for precise event and temporal control; achieved **22.62% exact match** on 420 MMAE examples, versus 13.57% for the strongest baseline.

Research Assistant | Info Lab, Worcester Polytechnic Institute

Advisor: Prof. Kyumin Lee

- Contributed to a two-stage framework for few-shot species identification from DNA melting curves, fusing BERT-encoded statistical prompts with time-series patch embeddings.
- Conducted model training, baseline comparisons, and ablation studies; achieved **93.08% mean accuracy** on 65 shark and ray species with only three training samples per species.

Research Assistant | King's Lab, Worcester Polytechnic Institute

Advisors: Prof. Benjamin Nephew & Prof. Dmitry Korkin

Alzheimer's Disease MRI Classification Project

- Developed 3D CNN-2D CNN-ViT hybrid architectures for early-stage Alzheimer's classification using the ADNI dataset.

PUBLICATIONS

Harmony: Bridging Audio Understanding and Editing through Agentic Collaboration.

Under review at TMLR | Project page

A Two-Stage Fusion Framework for Few-Shot Species Identification from Melting Curve Data

Chengqian Zhang, Kyumin Lee, and **Zhongyi Tong**.

ACM BCB 2026 | Paper PDF